



## Assignment: Engineering Intern

**Time: 24 hours**

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### The Problem

Millions of people generate data about themselves every day, what they do, when they do it, how often, and how it changes over time. Most of this data sits unused.

Build a backend system that takes a stream of user activity, finds patterns in it, and tells each user something genuinely useful about themselves that they could not see on their own.

That is the entire brief. What the activity is, what useful means, and how you build it - that is your problem to solve.

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### Constraints

- Your system must use both rule-based logic and an LLM not one or the other
  - Must accept real input and return real output
  - A reviewer should be able to run your system following your README
  - If a part is incomplete, document it explicitly what is missing, why, and what you would do to finish it
  - No prescribed stack, no prescribed data format, no prescribed number of endpoints
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### Submission

GitHub repository with working code, a README that runs, and a short note 200 words maximum on the single most important decision you made and was there any trade-off made.

**Deadline: 24 hours from receiving this.**